Carlton Knox

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Research Interests

Computer Architecture • GPU Architecture • High-Performance Computing • Performance Analysis

Education

University of Michigan — Ann Arbor, MI
Ph.D. in Computer Science and Engineering — 2026 – Present

Boston University, College of Engineering — Boston, MA
B.S. in Computer Engineering — 2019 – 2023
Magna Cum Laude
GPA: 3.85

Research Experience

Computer Engineering Lab, University of Michigan
PhD Student — Fall 2026 – Present
Advisor: Prof. Reetuparna Das

- Investigating hardware support for emulating FP32 on GPUs using lower-precision (e.g., BF16) multipliers, and how it maps onto systolic arrays and CUDA-core-style vector units.

Performance and Energy Aware Computing Lab, Boston University
Undergraduate Researcher — Jun 2021 – May 2022
Advisor: Prof. Ayse Coskun

- Developed machine-learning-based methodology for generating fine-grained CPU temperature maps from sparse on-die sensor data.
- Integrated benchmark data from x86 architectural simulators with PACT thermal simulation to train temperature prediction models.
- Built Python visualization tools to analyze time-series thermal behavior using seaborn/pandas.
- Research published at *ASME InterPACK 2022*.

Computer Architecture and Automated Design Lab, Boston University
Undergraduate Researcher — Spring 2023
Advisor: Prof. Martin Herbordt

- Contributed to research on OS scheduling for HPC applications, focusing on extracting useful work from cycles typically wasted on MPI synchronization.
- Implemented early algorithms for a library to schedule "extra work" during synchronization in load-imbalanced workloads.
- Research published at *2024 IEEE High Performance Extreme Computing Conference (HPEC)*.

Industry Experience

Advanced Micro Devices
Server Performance Engineer — 2023 – 2026

- Contributed to post-silicon bring-up of the Zen 5 architecture, characterizing memory bandwidth behavior across varied topology and core counts.
- Automated controlled memory bandwidth experiments to identify microarchitectural bottlenecks and uncover pre-release silicon bugs.
- Developed and maintained infrastructure for performance-monitoring and profiling tools across Zen5/Zen6 EPYC CPUs.
- Created a Python-based PMC automation tool for filtering, regrouping, and generating validated formulas for customer-facing performance metrics, reducing manual effort and improving accuracy.
- Built automation to streamline large-scale performance characterization, where a full run of 40+ benchmarks and sub-tests spanning HPC, AI/ML, SPEC CPU, database, and web/Java workloads takes several weeks per platform.
- Generalized the automation into a platform-agnostic layer supporting multiple EPYC CPU generations (Zen 4, Zen 4 X3D, Zen 5, Zen 5c, Zen 6), handling differing BIOS vendors and settings, frequency ranges, and topologies to enable 20+ tuning and sensitivity configurations per workload per platform, such as fixed-frequency and production settings.

Server Performance Engineering Intern Summer 2022

- Analyzed EPYC CPU performance using Linux Perf and eBPF.
- Characterized kernel/user call-stack behavior across high-performance workloads to identify performance anomalies.

Publications

- Cycle-Stealing in Load-Imbalanced HPC Applications, P. H. Chen, A. Bali, S. Yang, P. Haghi, **C. Knox**, B. Li, A. A. Abouelmagd, A. Skjellum, and M. Herbordt, in *2024 IEEE High Performance Extreme Computing Conference (HPEC)*, Wakefield, MA, USA, 2024, pp. 1–8. Outstanding Student Paper Award.
- Machine Learning and Simulation-Based Temperature Prediction on High-Performance Processors, **C. Knox**, Z. Yuan, and A. K. Coskun, in *ASME 2022 International Technical Conference on Packaging and Integration of Electronic and Photonic Microsystems*, Garden Grove, CA, USA, Oct. 2022.
- PACT: An Extensible Parallel Thermal Simulator for Emerging Integration and Cooling Technologies, Z. Yuan, P. Shukla, S. Chetoui, **C. Knox**, S. Nemtzow, S. Reda, and A. Coskun, in *31st ACM SIGDA University Demonstration at DAC 2021*, Dec. 2021. University Demo Best Demonstration, Third Place.
- Towards Fast and Accurate Parallel Chip Thermal Simulations with PACT, Z. Yuan, P. Shukla, S. Chetoui, **C. Knox**, S. Nemtzow, S. Reda, and A. K. Coskun, *WOSET 2021.*

Teaching Experience

Boston University

Teaching Assistant - Operating Systems Spring 2023

- Guided students on implementing OS primitives, including pthreads and a basic filesystem.
- Supported students through debugging of complex system-level projects.

Teaching Assistant - Computer Organization Fall 2022

- Helped students understand and implement basic CPU modules in verilog.
- Assisted in debugging, lab demonstrations, and assignment evaluations.

Selected Projects

Optimizing Graph Algorithms for Multicore CPU 2022

- Implemented parallel graph traversal algorithms using OpenMP and Pthreads, including dynamic load-balancing
- Demonstrated performance uplift with realistic test graphs

Tetris Implementation on FPGA 2022

- Implemented Tetris game state logic (piece rotation, collision detection) entirely in Verilog.
- Created a UART-based FPGA→Arduino interface for streaming frame data to a 32×64 RGB matrix display.

Cache Replacement Policy in gem5 2022

- Implemented Adaptive Replacement Cache (ARC) in gem5 simulator and evaluated interactions with range of cache sizes and benchmark patterns.

Other Activities

BU High Performance Computing Club 2019–2023
President

- Organized HPC workshops and led BU's participation in student cluster competitions.

SC21 & SC22 Student Cluster Competitions

- Competed in large-scale HPC benchmarks; contributed to performance tuning on GPU clusters.
- Achieved highest HPL score in SC22.

Society of Hispanic Professional Engineers (SHPE) 2019–2023
Treasurer

- Managed club finances and secured funding for professional development events.

Skills

Programming: C, Python, Verilog, Bash, Pthreads
HPC: MPI, OpenMP
Tools/Technologies: Linux Perf, eBPF, Pandas, gem5, FPGA
Languages: English, Spanish